One Seed, Many Friends

A short mathematical note on Jan Mikulík's finite causal Shannon theorem

With a little help from my friends.

One line, many crossings. The image recalls Wheeler's conjectural picture, recounted in Feynman's Nobel lecture: many electrons imagined as crossings of a single worldline. Our parallel concerns the word *one*. In information theory we can ask a precise question: how much randomness must be drawn once so that a simulator can answer whichever questions its friends later choose? No statement about the number of electrons follows from the information theorem.

The quantifiers matter. Fix a finite tree T of full histories, with an outcome law for each admissible action at each nonterminal node. A deterministic adaptive policy π chooses its next action from the history it has actually seen. An exact causal simulator draws a seed S once and answers by $D(S, h, a)$. The same distribution of S and the same decoder D must work for every policy. In particular, if two friends have the same history and take the same action, their replies coincide *for each fixed seed*. Their next questions can still lead them down different branches.

Finite causal Shannon theorem (Mikulík, Theorem 1.1). Let X_π be the random terminal transcript with law P_π, put $I_\pi = -\log_2 P_\pi(X_\pi)$, and define

$$F_*(t) = \min_\pi \Pr(I_\pi \le t), \qquad L(T) = \int_0^\infty (1 - F_*(t))\, dt.$$

Writing $C(T)$ for the minimum Shannon entropy of an exact causal seed, the theorem gives a finite seed S_G with

$$\boxed{L(T) \le C(T) \le H(S_G) \le L(T) + \frac{1 + \log_2 e}{2}.}$$

The additive constant is about 1.22135 bits and has prior coupling-theoretic provenance. The distinctive requirement here is one decoder that agrees at every shared history. The quantity being bounded is seed entropy, not storage for the decoder or its running time.

Why the lower bound holds. For any positive-mass seed atom s, let $x_\pi(s)$ be its transcript. Since this transcript is a deterministic function of s, $P_\pi(x_\pi(s)) \ge \Pr(S = s)$. Consequently $J = -\log_2 \Pr(S = s)$ evaluated at the sampled atom dominates transcript surprisal pointwise for every policy. Thus $\Pr(J \le t) \le F_*(t)$. Integrating the complementary distribution gives $H(S) \ge L(T)$. The simultaneous upper bound is the causal greedy construction in the cited manuscript; it does not follow merely by coupling the terminal laws.

Proposition. *Two friends can have identical transcript laws and nevertheless be coupled in a way that no shared causal decoder can realize.*

Proof. Both friends first ask the same question and must receive a fair bit. They then choose different actions, each with deterministic output zero. If B is fair, assign first replies B and $1-B$ to the two friends. Each friend's transcript has exactly the prescribed law, yet on every atom the first replies disagree. A causal decoder at the common root would have to give both replies at once. Conversely, giving both friends B is an exact simulator for the full tree. One bit suffices and is necessary because the first reply is fair. In this example $L(T) = C(T) = 1$. □

My friend need not ask my next question.
It is enough that we remember the same answer to the first.

Sources. J. Mikulík, *One Seed for Every Adaptive Policy: Exact Causal Simulation and Sharp Entropy Asymptotics*, 13 October 2026, Theorem 1.1; companion repository at the manuscript revision. S. Compton et al., *Minimum-Entropy Coupling Approximation Guarantees Beyond the Majorization Barrier*, AISTATS 2023. R. P. Feynman, Nobel lecture, 1965 (historical image, not a premise).